Yirong Ding

Ph.D. Student | Myers-Lawson School of Construction | Virginia Tech
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RESEARCH INTERESTS

- Disaster resilience and infrastructure service equity • Geospatial and network analysis
- Explainable AI for the built environment • Knowledge graphs and multi-agent systems

TECHNICAL SKILLS

Programming and Analytics: Python, R, Java, data analysis, machine learning, deep learning, explainable AI, causal discovery, spatio-temporal analysis, geospatial analytics
AI and Data Systems: Knowledge graphs, grounded LLMs, multi-agent systems
Software: ArcGIS, SAP2000, ETABS, Midas, Revit, Dynamo, Navisworks, ArchiCAD, Tekla, AutoCAD

EDUCATION

Virginia Tech — **Blacksburg, Virginia**
- Ph.D. in Environmental Design and Planning — Expected May 2027
  Dissertation: Towards Socio-Physical Resilience Geospatiotemporal Knowledge Graphs and Multi-Agent LLMs for Grounded Disaster Decision Support
  Advisor: Dr. Lu Zhang; Co-advisor: Dr. Yang Zhang
- M.Eng. in Computer Science (Simultaneous Degree) — Expected December 2026
- Future Professoriate Graduate Certificate — 2026

Hong Kong University of Science and Technology — **Hong Kong, China**
- M.Sc. in Civil Infrastructural Engineering and Management — December 2023
  Thesis: Low-cost and Accurate Occupancy Prediction on Campus
  Advisor: Dr. Zhe (Walter) Wang

Chongqing University — **Chongqing, China**
- B.Eng. in Civil Engineering — June 2016

ACADEMIC APPOINTMENTS

Myers-Lawson School of Construction, Virginia Tech — **Blacksburg, Virginia**
Graduate Research Assistant — **01/2024–Present**
Graduate Teaching Assistant — **Fall 2024; Spring, Summer and Fall 2026**

RESEARCH EXPERIENCE

Community-Centric Lifeline Accessibility during Hurricane Helene — **09/2025–Present**
- Constructed a statewide directed road network and measured daily access to seven lifeline-facility types for 2,975 census block groups using 30 days of observed road closures.
- Developed spatial hurdle and interpretable spatio-temporal models to distinguish isolation occurrence, severity, persistence, and changing recovery-stage drivers.

Knowledge Graph and Multi-Agent AI System — **03/2025–Present**
- Built a working geospatiotemporal knowledge-graph and multi-agent LLM prototype linking Hurricane Helene road, power, facility, community, and policy evidence.
- Designed source-linked retrieval and reasoning so proposed responses can be traced to underlying records; formal evaluation of accuracy, citation faithfulness, and decision quality is planned.

Flood Hazard Maps and Local Flood Policy in the United States — **02/2024–Present**
- Examined flood-map coverage, currency, uncertainty, policy barriers, and agreements and discrepancies between FEMA and First Street Foundation delineations.
- Research supported by Virginia Tech's +Policy Fellows Program.

Context-Sensitive Disaster Resilience and Equity Analysis — **08/2024–08/2025**
- Used explainable geospatial machine learning on 2,836 census tracts affected by Hurricanes Irma and Ian to study impact containment, resource mobilization, and recovery capability.
- Identified time- and place-dependent links among hazard exposure, building conditions, socioeconomic capacity, resilience, and equity.

- Research supported by Virginia Tech's ISCE Scholars Program.

Public Infrastructure Service Inequality during Disasters **01/2024–01/2025**

- Studied 311 solid-waste service requests in Lee County, Florida, during Hurricanes Irma and Ian using event study, LSTM forecasting, spatial autocorrelation, and emerging hotspot analysis.
- Quantified demand and resolution-time disparities across inland, coastal, and vulnerable communities.

Low-Cost and Accurate Occupancy Prediction on Campus **02/2023–12/2023**

- Developed Wi-Fi-based occupancy models using random forests, DNN, LSTM, Transformer, and XGBoost for scalable campus space-utilization analysis.

PUBLICATIONS AND MANUSCRIPTS

Peer-Reviewed Journal Articles

- Yang Zhang*, **Yirong Ding**, Lu Zhang, Jennifer L. Irish, Yang Shao. Flood Hazard Maps for Local Flood Policy in the United States: Barriers and Opportunities. *Journal of the American Planning Association*, 1–16, 2026.
- **Yirong Ding**, Lu Zhang*, Yang Zhang. Context-Sensitive Analysis of Disaster Resilience and Equity through Geospatial Explainable Machine Learning. *Sustainable Cities and Society*, 139, 107203, 2026. [2025 Journal Impact Factor: 13.3]
- **Yirong Ding**, Lu Zhang*, Yang Zhang, Christopher W. Zobel. Spatial–Temporal Dynamics of Public Infrastructure Service Inequality during Disasters. *Natural Hazards Review*, 27(1), 04025067, 2026.

Manuscript Under Review

- **Yirong Ding**, Yang Zhang*, Jennifer L. Irish, Lu Zhang, Yang Shao. When Flood Maps Disagree: Regulatory and Alternative Flood Hazard Assessment in Local Flood Governance. *Under Review*.

Manuscripts in Preparation

- **Yirong Ding**, Yang Zhang*, Jennifer L. Irish, Lu Zhang, Yang Shao. Place-based Uncertainties: Contextualizing Agreement and Divergence between Regulatory and Alternative Flood Hazard Assessments for Local Flood Governance. *In Preparation*.
- **Yirong Ding**, Lu Zhang*, Yang Zhang. Interpretable Spatio-Temporal Modeling of Lifeline Facility Accessibility during Hurricane Helene: A Neural Additive Mixture-of-Experts Approach. *In Preparation*.
- **Yirong Ding**, Lu Zhang*, Yang Zhang. PowerKG-Harness: An Auditable Knowledge Graph-Grounded LLM System for Facility-Level Power Outage Decision Support. *In Preparation*.

Conference Proceedings

- **Yirong Ding**, Lu Zhang*, Yang Zhang. Understanding Context-Sensitive Relationships between Disaster Resilience and Equity through Geo-Information Embedded Explainable Machine Learning. *Construction Research Congress 2026: Sustainable and Resilient Infrastructure Systems and Materials Engineering and Design*, pp. 358–367. American Society of Civil Engineers, Reston, VA. https://doi.org/10.1061/9780784487006.036. Best Paper per CRC Track (Track 8).
- **Yirong Ding**, Lu Zhang*, Yang Zhang. Characterizing Spatial and Temporal Heterogeneity in Lifeline Facility Accessibility in Hurricane Helene. *2026 ASCE International Conference on Computing in Civil Engineering (i3CE)*. Accepted; invited to submit an extended version to a *Journal of Computing in Civil Engineering* special collection.
- **Yirong Ding**, Lu Zhang*, Yang Zhang, Christopher W. Zobel. Public Infrastructure Service Disparities: Spatiotemporal Analysis of Solid Waste Service Requests during Hurricane Irma. *Computing in Civil Engineering 2025: Resilient, Robotic, and Educational Systems*, pp. 339–349. American Society of Civil Engineers, Reston, VA. https://doi.org/10.1061/9780784486443.038.

SELECTED PRESENTATIONS

- **Yirong Ding**, Yang Zhang*, Jennifer L. Irish, Lu Zhang, Yang Shao. Assessing Agreement and Divergence in FEMA and FSF Flood Hazard Delineations. *Virginia Crossroads Flood Resilience Symposium*, Charlottesville, Virginia, 2026.
- **Yirong Ding**, Lu Zhang*, Yang Zhang. Understanding Context-Sensitive Relationships between Disaster Resilience and Equity through Geo-Information Embedded Explainable Machine Learning. *ASCE Construction Institute and Construction Research Congress Joint Conference*, San Antonio, Texas, 2026. Best Paper per CRC Track, Track 8.
- **Yirong Ding**, Lu Zhang*, Yang Zhang. Towards Socio-Physical Resilience: Geospatiotemporal Knowledge Graphs and Multi-Agent LLMs for Traceable Disaster Decision Support. Poster, *2026 ASCE International Conference on Computing in Civil Engineering (i3CE)*, Incheon, South Korea.

- **Yirong Ding**, Lu Zhang*, Yang Zhang. Characterizing Spatial and Temporal Heterogeneity in Lifeline Facility Accessibility in Hurricane Helene. Presentation, *2026 ASCE International Conference on Computing in Civil Engineering (i3CE)*, Incheon, South Korea.
- Yang Zhang*, **Yirong Ding**, Lu Zhang, Jennifer L. Irish, Yang Shao. Flood Hazard Maps and Local Flood Policy in the U.S.: Barriers and Opportunities. *Natural Hazards Workshop*, 2025.

* Corresponding author.

PROPOSAL DEVELOPMENT

Impact-Based Coastal Flood Prediction for Virginia's Coastal Communities: A Knowledge-Driven Decision Support Framework **2026**

- Student PI and lead proposal writer; Virginia Water Resources Research Center Student Grant Research Proposals; $10,000 requested; not funded.

Rapid Flood Impact and Recovery Assessment and Knowledge-Based Decision Support System **2025**

- Student PI and lead proposal writer; Virginia Water Resources Research Center Student Grant Research Proposals; $10,000 requested; not funded.

Faculty-led proposal support **2024–2026**

- Supported faculty-led proposals for NSF Fire Science Innovations through Research and Education (FIRE) and EDU Core Research (ECR:Core) by preparing references; developed overall framework figures for other proposals.

TEACHING EXPERIENCE

Virginia Tech **Blacksburg, Virginia**
Graduate Teaching Assistant and Course Instructional Support **Spring 2024–Present**

- Supported ten course sections with 31–61 students per section and 475 total enrollments.
- Assessed exercises, homework, quantity-takeoff and estimating work, workshop submissions, and business-plan deliverables using instructor-provided rubrics.
- Provided explanatory and encouraging feedback, held two office-hour sessions weekly, maintained routine student communication, and coordinated approved testing accommodations.
- Assisted with team formation and project-presentation logistics and participated in presentation evaluation.

CEM 3084 – Construction Economy | Instructor: Dr. Lu Zhang **7 sections; 352 enrollments**

- Topics supported: time value of money, comparison of alternatives, financing, depreciation, financial statements, construction accounting, project cash flows, and business-plan development.

BC 2024 – Construction Principles II | Instructor: Dr. Xuan Lv **1 section; 40 enrollments**

- Fall 2024; quantity takeoff and estimating for masonry, wood framing, roofing, interior finishes, and related workshop submissions.

BC 2014 – Construction Principles I | Instructor: Dr. Xuan Lv **2 sections; 83 enrollments**

- Summer 2026 and Fall 2026; critical path method, square-foot estimating, foundation quantity takeoff and pricing, foundation reinforcement, and weekly exercises.

TEACHING PROFESSIONAL DEVELOPMENT

Future Professoriate Graduate Certificate, Virginia Tech **2026**

- Completed Preparing the Future Professoriate, Contemporary Pedagogy, and Diversity for a Global Society.
- Developed preparation in higher-education systems, contemporary pedagogy, inclusive learning environments, and communication across cultural and disciplinary backgrounds.

GRAD 5004 – GTA Training Workshop, Virginia Tech **Fall 2024**

- Completed university GTA training through workshop and webinar sessions.

MENTORING

ASCE VIMS Datathon Undergraduate Project Co-Mentor **2026**

- Co-mentored two undergraduate students with Dr. Tianzhi He on 3D Gaussian Splatting reconstruction of high-fidelity indoor built-environment models.
- Coordinated workflow integration, clarified data formats and methods, calibrated expectations to student preparation, and supported completion of the project submission.

HONORS AND AWARDS

- Best Paper per CRC Track, Track 8: Sustainable and Resilient Infrastructure Systems and Materials Engineering and Design, CI & CRC Joint Conference — 2026
- Research Award, Myers-Lawson School of Construction, Virginia Tech — 2025
- Phi Kappa Phi Member, National Honor Society of Phi Kappa Phi — 2025
- Disaster Resilience Interdisciplinary Graduate Education Program (DR-IGEP) — 2025
- Pratt Fellowship, Virginia Tech — 2024
- Excellent League Member, Yunnan Design Institute Group Co., Ltd. — 2019, 2022
- Outstanding Graduation Project, Chongqing University — 2016
- Bronze Medal, AkzoNobel China Student Sustainability Award — 2014

ACADEMIC AND UNIVERSITY SERVICE

- Student Member, ASCE Visualization, Information Modeling, and Simulation Committee — 06/2026–Present
- Student Member, ASCE Global Center for Excellence in Computing — 05/2025–Present
- Student Member, ASCE Data Sensing and Analysis Committee — 05/2025–Present
- Conference Reviewer: ASCE i3CE 2025/2026; CI & CRC Joint Conference 2026
- Conference Reviewer: CONVR/ICIO 2026; SASBE/FR/BDTSC 2026 Joint Conference
- Journal Reviewer: *Computers, Environment and Urban Systems*
- Graduate Ambassador, Myers-Lawson School of Construction, Virginia Tech

PROFESSIONAL EXPERIENCE

Yunnan Design Institute Group Co., Ltd. — **Yunnan, China**
Structural Design Engineer / Design Supervisor / Design Representative — **04/2017–01/2023**

- Conducted or led structural calculations, seismic analysis, and design for approximately 30 building projects totaling more than 220,000 m².
- Led design management and coordination as design general contractor for the RMB 1.618 billion Fuxian Lake Vegetation Restoration PPP Project.
- Resolved on-site design issues for five building projects totaling more than 680,000 m².

Hangxiao Steel Structure Co., Ltd. — **Zhejiang, China**
Steel Structure Engineer and BIM Engineer — **07/2016–03/2017**

- Designed Tekla connections for four building projects with investments exceeding RMB 1 billion.
- Completed BIM modeling for the 24,363.59 m² Huaxia Yufu commercial project in Kunming.

Kunming Survey, Design and Research Institute Co., Ltd. of CREEC — **Yunnan, China**
Structural Design Intern — **01/2015–02/2015**

- Studied structural design and analysis for Kunming Metro Line 3 subway stations and sections.

SELECTED EARLIER RESEARCH AND DESIGN PROJECTS

Multi-Specialty Joint Graduation Project: Chengdu Xinglong Lake Resort — **2016**

- Led the structural discipline's layout, calculations, and drawing design for a frame-shear-wall high-rise building and coordinated with architecture, MEP, and construction-economics teams.
- Project recognized as an Outstanding Graduation Project.

National Student Research Training Program: Infrared Detection of Concrete-Filled Steel-Tube Compactness — **Undergraduate**

- Led a three-member team, developed research methods and test plans, modified testing equipment, and completed the project with a Good completion rating.